Analysis of Variance

Overview

Continuation of hypothesis tests for multiple proportions using the chi-square statistic

- Randomization test using the chi-square statistic
- Parametric chi-square statistic test for goodness-of-fit

One-way analysis of variance (ANOVA) for comparing multiple means

- Running a one-way ANOVA
- If there is time: Randomization test using an F-statistic

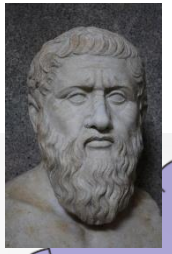
The big picture

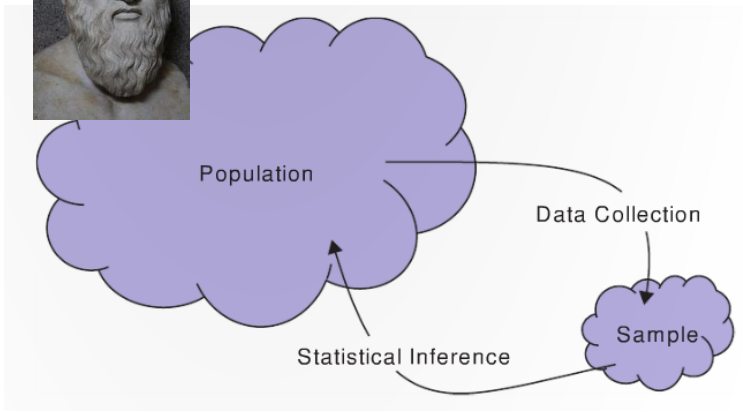
Review: The big picture of statistical inference

Statistic inference always involves using **samples** of data to make claims about a **population**

- i.e., using **statistics** to make claims about **parameters**

π, μ, σ, ρ





To quantify the **size of an effect**, we have created <u>confidence intervals</u>

- These intervals contain a parameter of interest most of the time

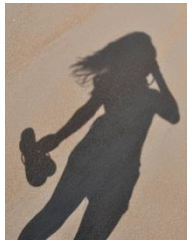
0.4 0.6

ρ

To assess if a particular **parameter value is plausible**, we have run <u>hypothesis tests</u>

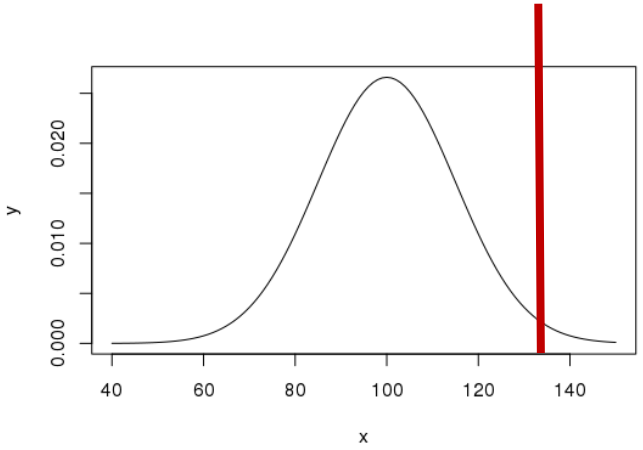
$H_0: \rho = 0$

$\hat{p}, \bar{x}, s, r$



Inference parametric probability models

For the past few weeks, we have used parametric probability functions for the null and sampling distributions

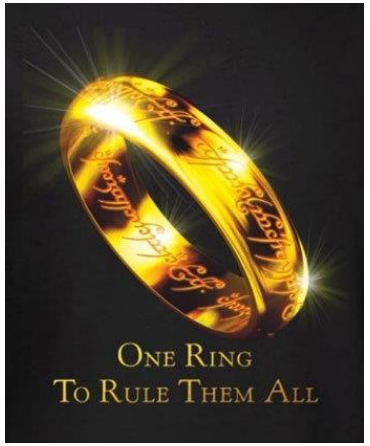


For confidence intervals, we will have formulas for SE estimates

- CI = stat ± q* · $\hat{SE}$ $\qquad \hat{SE} = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$

For hypothesis tests, we will have mathematical functions for the null distribution

- E.g., our null distribution will be a normal distribution



Data	1 Sample	2 Samples	> 2 Samples/Categories
Categorical data	H_0: $\pi = p_0$ H_A: $\pi \neq p_0$ <u>z-test</u> $z = \frac{\hat{p}-p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$	H_0: $\pi_1 = \pi_2$ H_A: $\pi_1 \neq \pi_2$ <u>z-test</u> $z = \frac{\hat{p_1} - \hat{p_2}}{\sqrt{\frac{\hat{p}(1-\hat{p})}{n_1} + \frac{\hat{p}(1-\hat{p})}{n_2}}}$	H_0: $\pi_1 = p_1$, $\pi_2 = p_2$, … , $\pi_k = p_k$ H_A: At least one p_i is different than specified <u>chi-square test</u> $\chi^2 = \sum_{i=1}^{k} \frac{(Observed_i - Expected_i)^2}{Expected_i}$ df = k - 1 Today!
Quantitative data	H_0: $\mu = v_0$ H_A: $\mu \neq v_0$ <u>One sample t-test</u> $t = \frac{\bar{x}-v_0}{s/\sqrt{n}}$ df = n - 1	H_0: $\mu_1 = \mu_2$ H_A: $\mu_1 \neq \mu_2$ <u>Two sample t-test</u> $t = \frac{\bar{x}_1-\bar{x}_2}{\sqrt{\frac{s_1^2}{n_1}+\frac{s_2^2}{n_2}}}$ df = min n_1 - 1, n_2 - 1	H_0: $\mu_1 = \mu_2 = \ldots = \mu_k$ H_A: At least one μ_i is different <u>Analysis of Variance</u> $F = \frac{\frac{1}{K-1}\sum_{i=1}^{K} n_i(\bar{x}_i - \bar{x}_{tot})^2}{\frac{1}{N-K}\sum_{i=1}^{K}\sum_{j=1}^{n_i}(x_{ij} - \bar{x}_i)^2}$ df_1 = k -1, df_2 = n - k Today!

Randomization test to compare multiple proportions

Testing more than two categories

A Yale professor was interesting in examining whether Yale students were equally likely to be born in each month

Q_1: What is the null and alternative hypotheses he could use to test this hypothesis?

Testing more than two categories

The hypotheses:

H_0: $\pi_{Jan} = \pi_{Feb} = \ldots = \pi_{Dec} = 1/12$

H_A: At least one of the proportions π_i is different

Chi-square goodness-of-fit test

If we want to test proportions for k > 2 categories, we can use:

1. A randomization test
2. A parametric "*chi-square goodness-of-fit*" test

These test:

H_0: $\pi_1 = a$, $\pi_2 = b$, … $\pi_k = z$

H_A: Some π_i is not as specified in H_0

The tests don't specify which proportion differs from what is specified in the null hypothesis, just that at least one proportion does differ

Chi-square statistic

The **chi-square statistic**, denoted χ^2, is found by comparing the **observed counts** from a sample with the **expected counts** derived from a null hypothesis and is computed as:

$$\chi^2 = \sum_{i=1}^{k} \frac{(\text{Observed}_i - \text{Expected}_i)^2}{\text{Expected}_i}$$

Note this is a Greek symbol even though it is a statistic ☹

Birth months from 198 Yale students

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	**Total**
Observed	20	17	21	24	11	18	12	17	13	14	17	14	**198**
Expected	16.5	16.5	16.5	16.5	16.5	16.5	16.5	16.5	16.5	16.5	16.5	16.5	**198**

Q: What is the expected value for each month? A: $1/12 \cdot 198$

$$\chi^2 = \sum_{i=1}^{k} \frac{(\text{Observed}_i - \text{Expected}_i)^2}{\text{Expected}_i} = 10.12$$

Q: Is the observed statistic beyond what we would expect if H_0 was true?

Creating a null distribution

For the Yale birth month example we can roll a die to create a null distribution…

How many sides would the die have?

- 12-sided die

What would the probability be of getting each side?

- 1/12

How many times would we roll the die to simulate one data set?

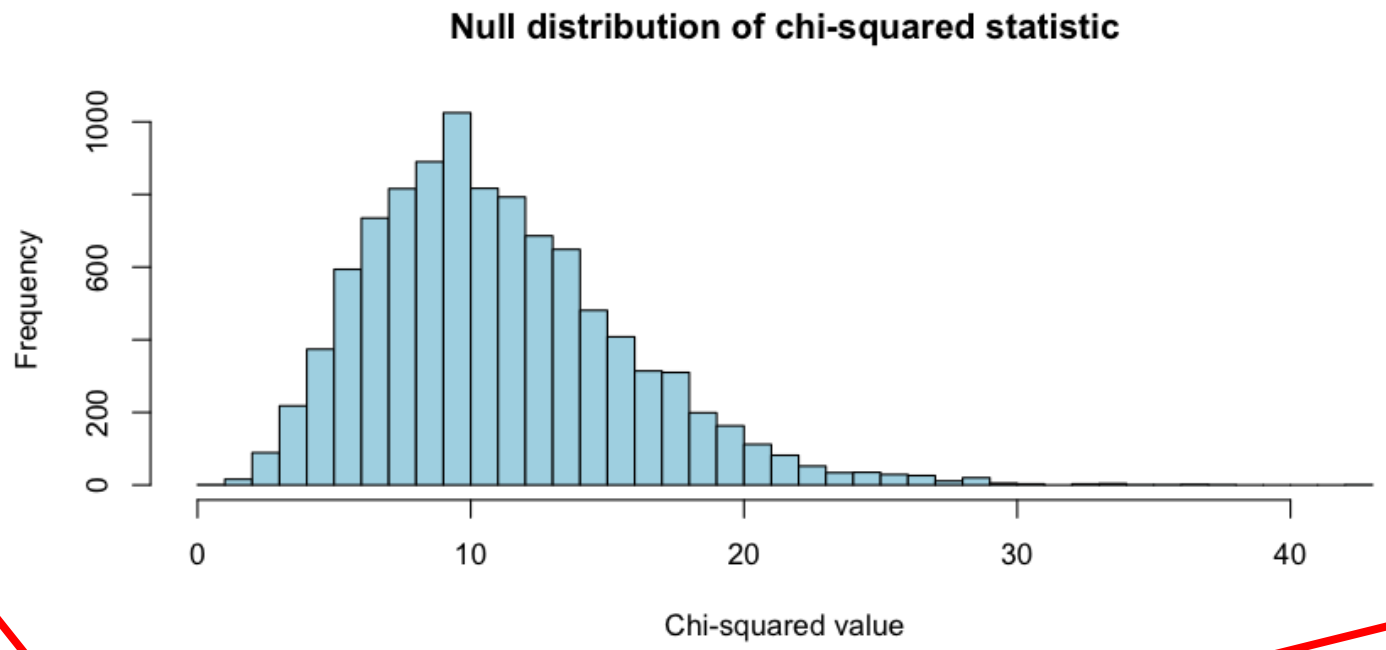
- n = 198
- Calculate χ^2 statistic based on these 198 rolls

$$\chi^2 = \sum_{i=1}^{k} \frac{(\text{Observed}_i - \text{Expected}_i)^2}{\text{Expected}_i}$$

How many times would we repeat this process?

- 10,000 times

Randomization null distribution



Vector of randomly generated counts in each category

"observed" data consistent with H_0

Vector of expected proportions for each category level

Vector length is the number of categories k

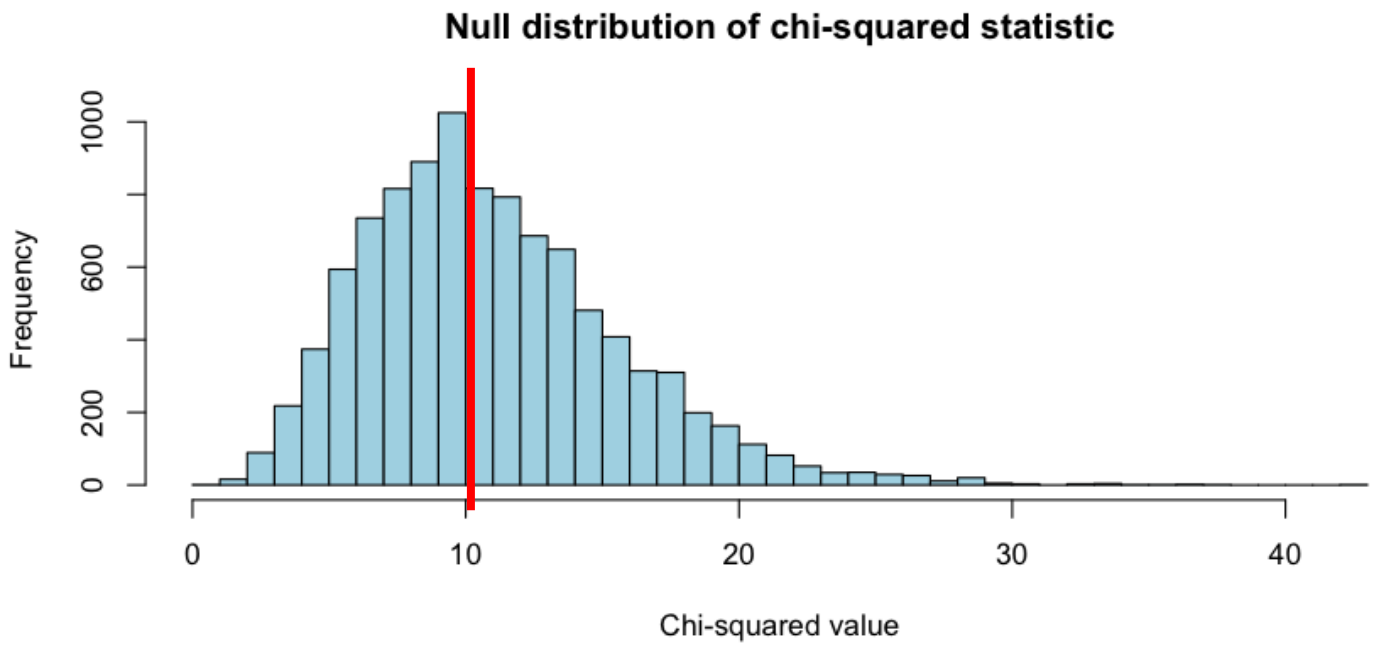
```
simulated_counts <- rroll(n,  expected_proportions)

simulated_counts <- rroll(198,  rep(1/12, 12))
```

Calculate statistic χ^2 and repeat 10,000 times

$$\chi^2 = \sum_{i=1}^{k} \frac{(\text{Observed}_i - \text{Expected}_i)^2}{\text{Expected}_i}$$

Randomization null distribution

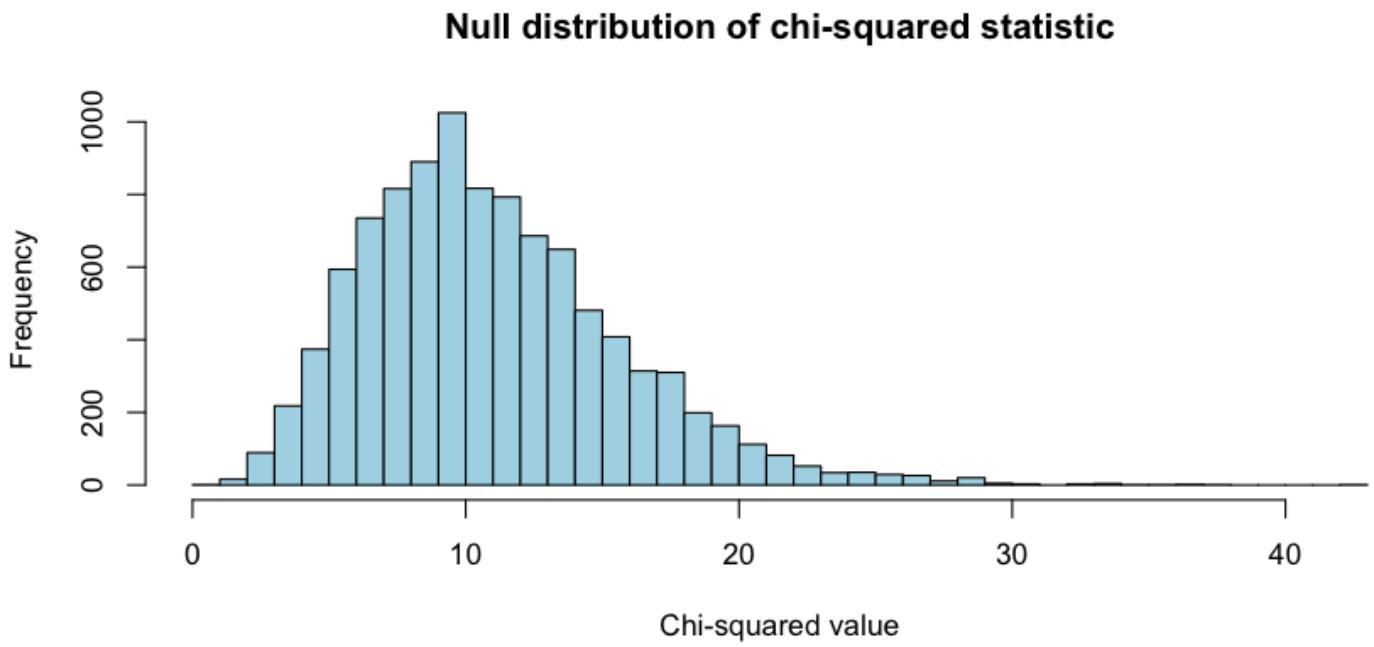


$\chi^2 = 10.12$

p-value = …?

χ^2 test for goodness-of-fit to compare multiple proportions

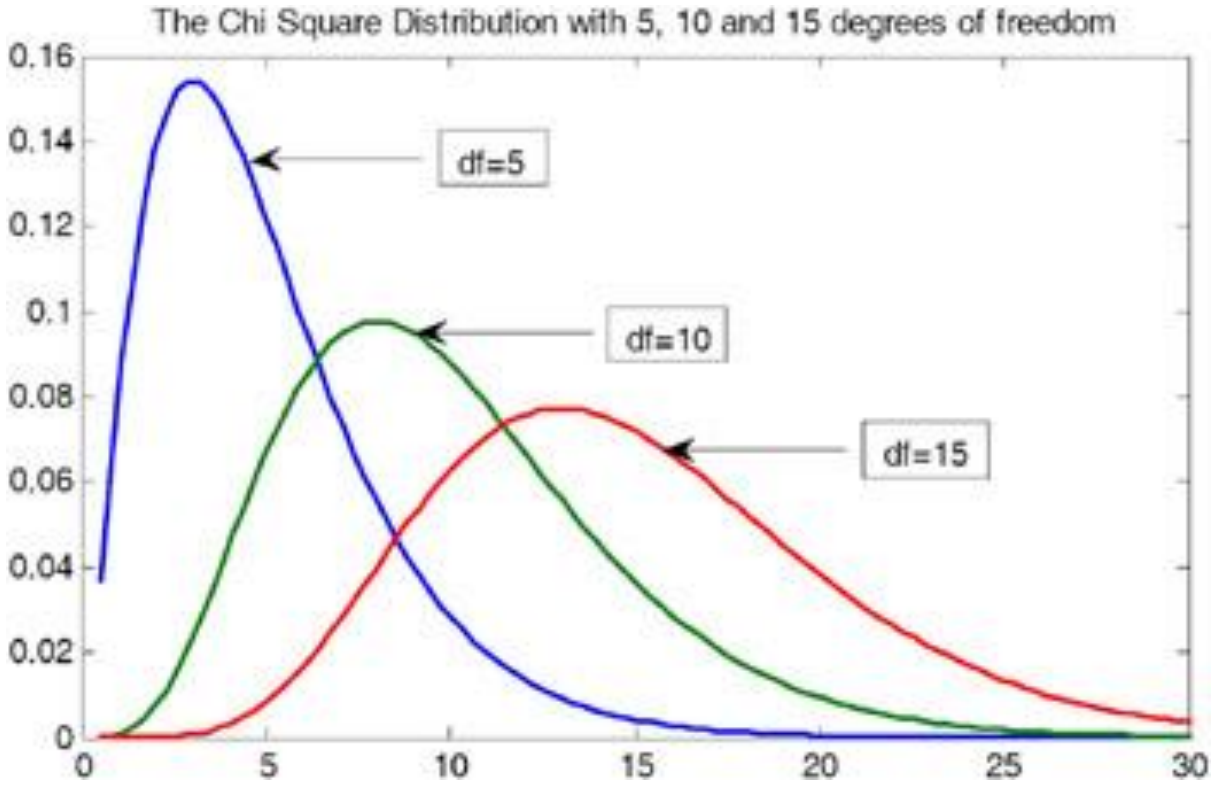
Parametric null distribution



Is there a parametric null distribution we could use instead of simulations?

Yes! The χ^2 distribution!

Chi-square distribution

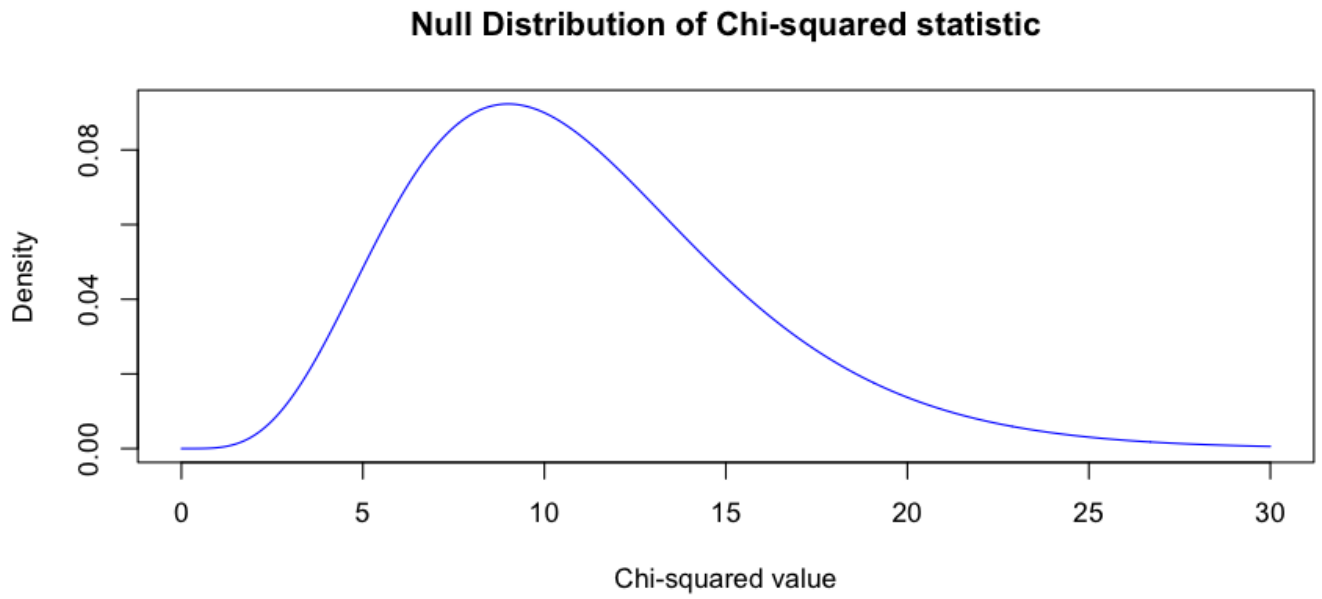


k is the number of groups

The χ^2 distribution has one parameter called 'degrees of freedom', which is equal to k - 1

χ^2 distribution can be used as a null distribution as long as there are at least 5 expected counts in every condition

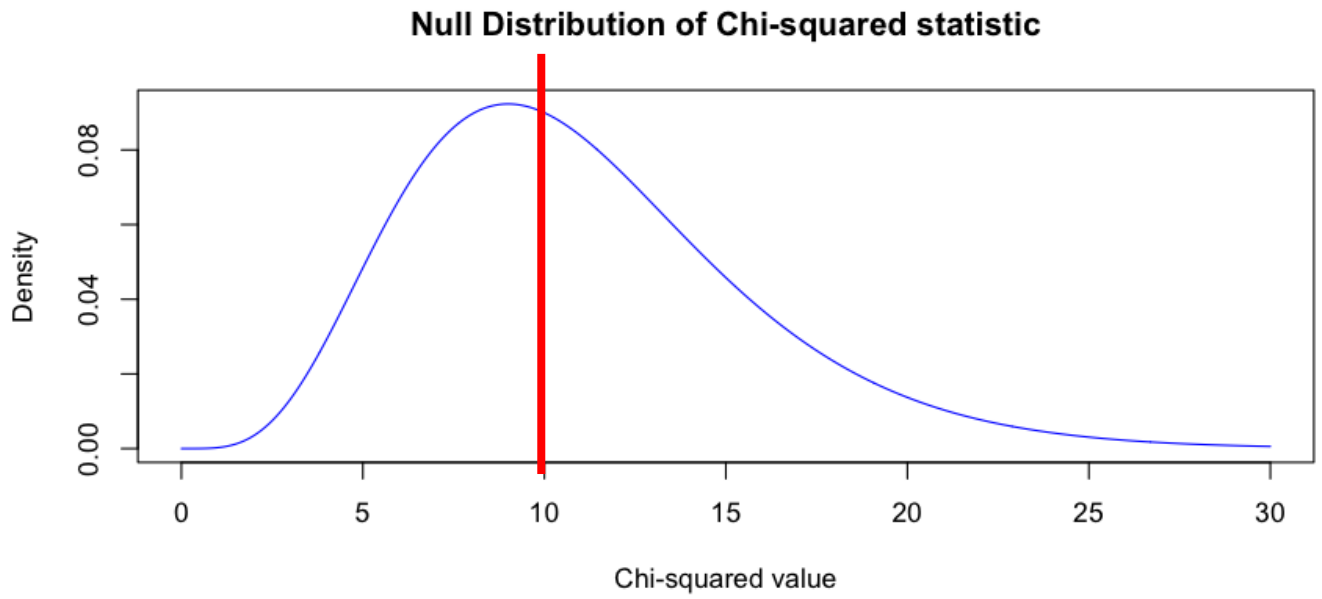
Chi-square distribution



To plot the chi-squared density we can use: y_vals <- dchisq(x_vals, df)

To get a p-value we can use: pchisq(chi_stat, df, lower.tail = FALSE)

Chi-square distribution



For Yale birth month example, since there 12 months there are 11 degrees of freedom (df = 11)

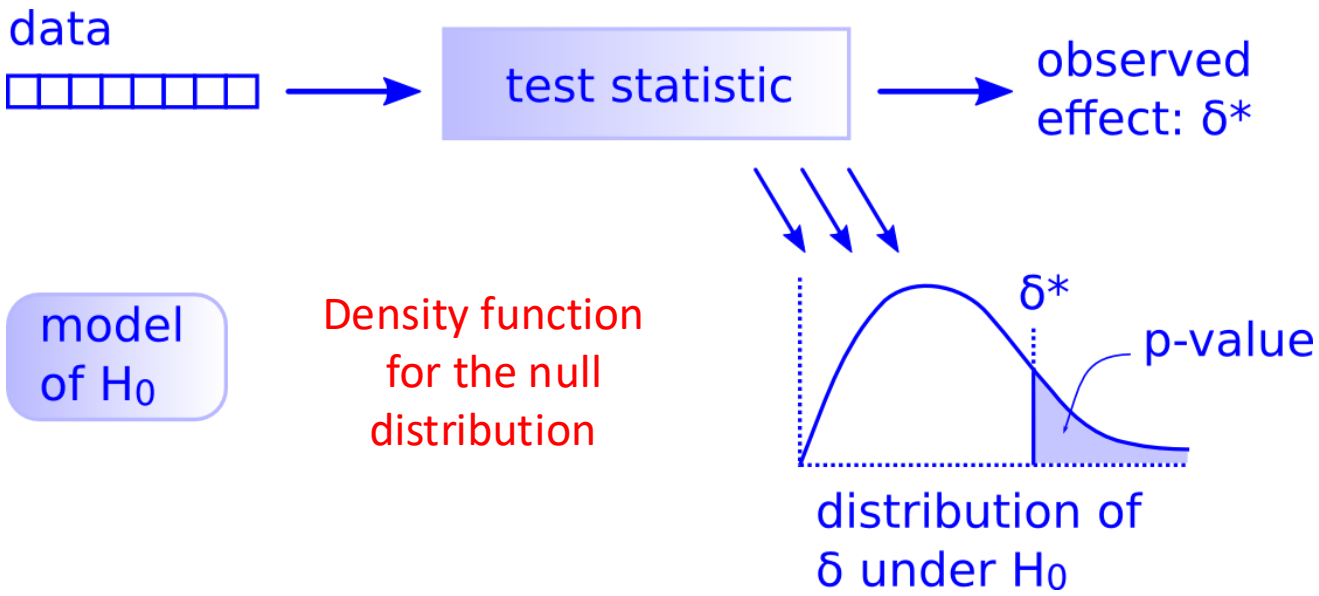
P-value based on the chi-squared distribution = …?

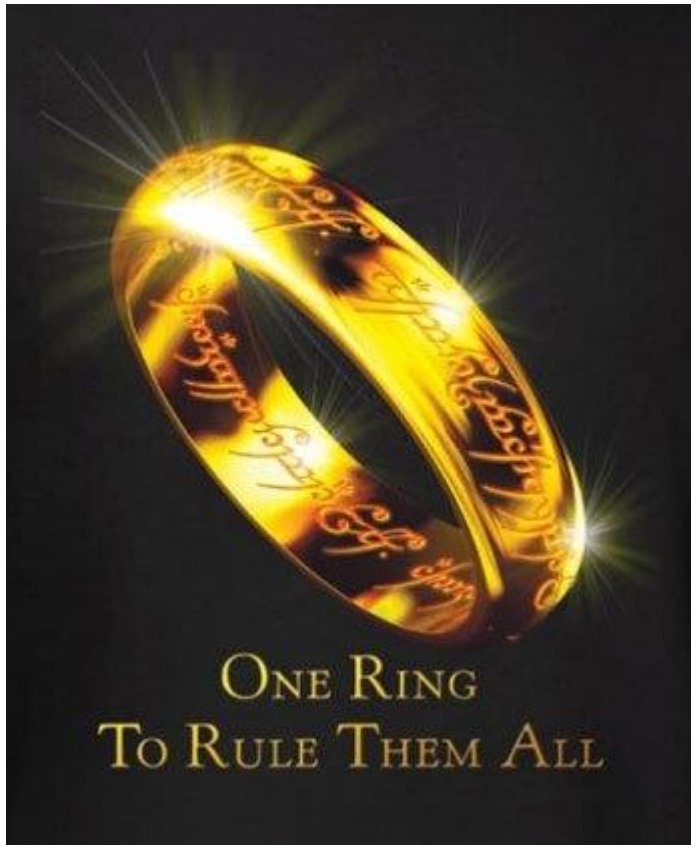
One-way analysis of variance (ANOVA):

Parametric test for comparing more than two means

One test to rule them all

There is only one hypothesis test!





Just follow the 5 hypothesis tests steps!

One-way ANOVA

An Analysis of Variance (ANOVA) is a parametric hypothesis test that can be used to examine if a set of means are all the same

H_0: $\mu_1 = \mu_2 = \ldots = \mu_k$

H_A: $\mu_i \neq \mu_j$ for some i, j

The statistic we use for a one-way ANOVA is the F-statistic

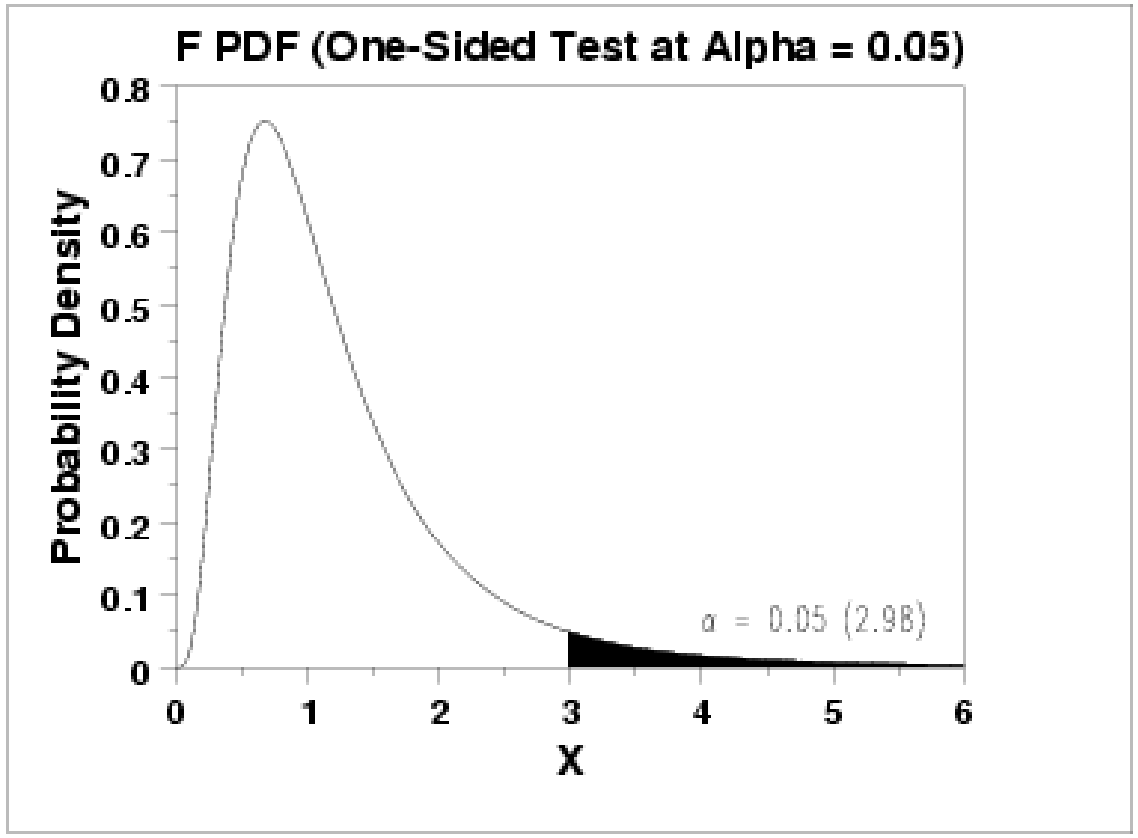
$$F = \frac{\text{between-group variability}}{\text{within-group variability}} = \frac{\frac{1}{K-1}\sum_{i=1}^{K} n_i(\bar{x}_i - \bar{x}_{tot})^2}{\frac{1}{N-K}\sum_{i=1}^{K}\sum_{j=1}^{n_i}(x_{ij} - \bar{x}_i)^2}$$

One-way ANOVA – the central idea

If H_0 is true, the F-statistic we compute from our data will come from an F-distribution if these conditions are met:

- The data in each group should follow a normal distribution
- The variances in each group should be approximately equal

We can get a p-value by finding the probability we will get a F-statistic larger than the observed F-statistic

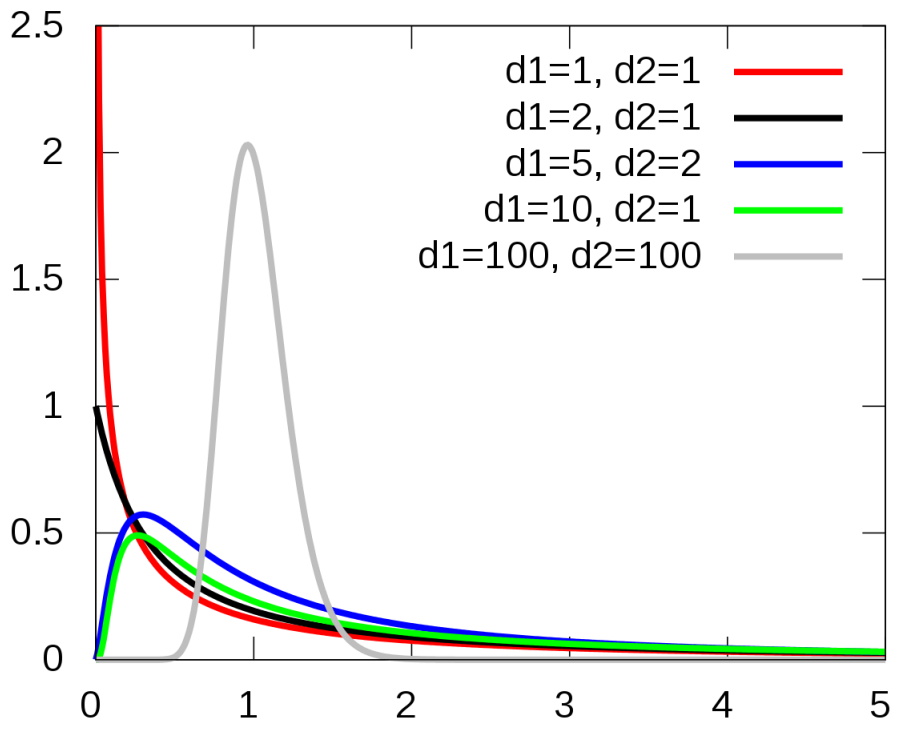


One-way ANOVA – the central idea

The F-distribution is a family of distributions that have two parameters: df_1 and df_2

When using the F-distribution as a null distribution for our F-statistic, the appropriate parameter values are:

- $df_1 = K - 1$
- $df_2 = N - K$



Comparing more than two means

A group of Hope College students wanted to see if there was an association between a student's major and the time it takes to complete a small Sudoku-like puzzle

	5	3	2		7			8
6		1	5					2
2			9	1	3		5	
7	1	4	6	9	2			
	2						6	
			4	5	1	2	9	7
	6		3	2	5			9
1					6	3		4
8			1		9	6	7	

Comparing more than two means

A group of Hope College students wanted to see if there was an association between a student's major and the time it takes to complete a small Sudoku-like puzzle

They grouped majors into four categories

- Applied science (as)
- Natural science (ns)
- Social science (ss)
- Arts/humanities (ah)

What is the first step of hypothesis testing?

1. State the null and alternative hypotheses

$\mathbf{H_0}$: $\mu_{as} = \mu_{ns} = \mu_{ss} = \mu_{ah}$

$\mathbf{H_A}$: $\mu_i \neq \mu_j$ for one pair of fields of study

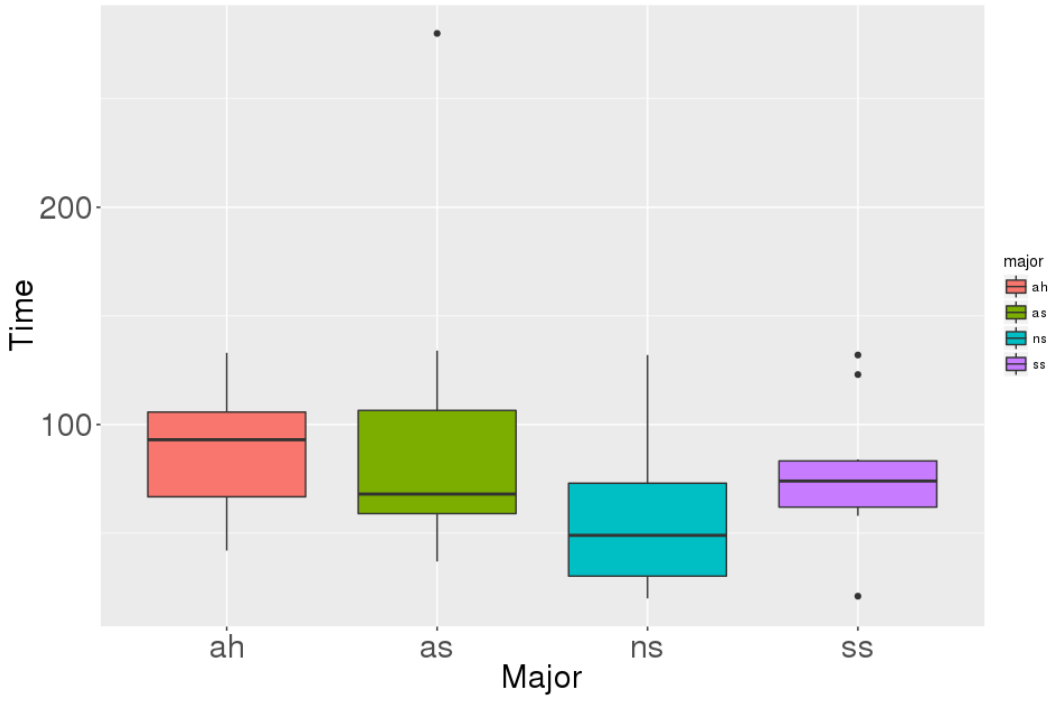
What should we do next?

Let's check if the ANOVA conditions are met first…

Checking ANOVA conditions ('assumptions')

We can check if the data in each group is relatively normal by creating boxplots and seeing:

- Is the data very skewed?
- Are there are many outliers?



We can check the equal variance condition by seeing if the ratio of the largest to smallest standard deviation is greater than 2

- $s_{max}/s_{min} < 2$

$s_{ah} = 27.9$

$s_{as} = 71.59$

$s_{ns} = 34.39$

$s_{as} = 31.89$

2. Calculating the observed F-statistic

$$F = \frac{\text{between-group variability}}{\text{within-group variability}} = \frac{\frac{1}{K-1}\sum_{i=1}^{K} n_i(\bar{x}_i - \bar{x}_{tot})^2}{\frac{1}{N-K}\sum_{i=1}^{K}\sum_{j=1}^{n_i}(x_{ij} - \bar{x}_i)^2}$$

K: the number of groups

N: total number of points

$\bar{x}_{tot}$: the mean across all the data

$\bar{x}_i$: the mean of group i

n_i: the number of points in group i

x_{ij} : the jth data point from group i

K = 4 different majors here

N = 40 total students in the full data set

$\bar{x}_{tot}$: the mean across Sudoku times

$\bar{x}_i$: the means for ah, as, ns, and ss

n_i = 10 students in each major

x_{ij} : the jth student's time from the ith major

2. Calculating the observed F-statistic

$$F = \frac{\text{between-group variability}}{\text{within-group variability}} = \frac{\frac{1}{K-1}\sum_{i=1}^{K} n_i(\bar{x}_i - \bar{x}_{tot})^2}{\frac{1}{N-K}\sum_{i=1}^{K}\sum_{j=1}^{n_i}(x_{ij} - \bar{x}_i)^2}$$

Fortunately, there is a function in the SDS1000 package that will calculate this for us!

get_F_stat(data_vector, grouping_vector)

- data_vector: a vector of quantitative data
- grouping_vector: a vector indicating which group the quantitative data is in

Let’s try this analysis in R…

```
# get the data
library(SDS1000)
sudoku_data <- read.table("MajorPuzzle.txt", header = TRUE)

# Extract vectors from the data frame  (how do we do this?)
completion_time <- sudoku_data$time
major <- sudoku_data$major
```

Let’s try this analysis in R…

We can get the F statistic using the get_F_stat() function

get_F_stat(data_vector, grouping_vector)

- data_vector: a vector of quantitative data
- grouping_vector: a vector indicating which group the quantitative data is in

Can you get the F statistic for the sudoku data?

```
obs_stat <- get_F_stat(completion_time, major)
```

= 1.370

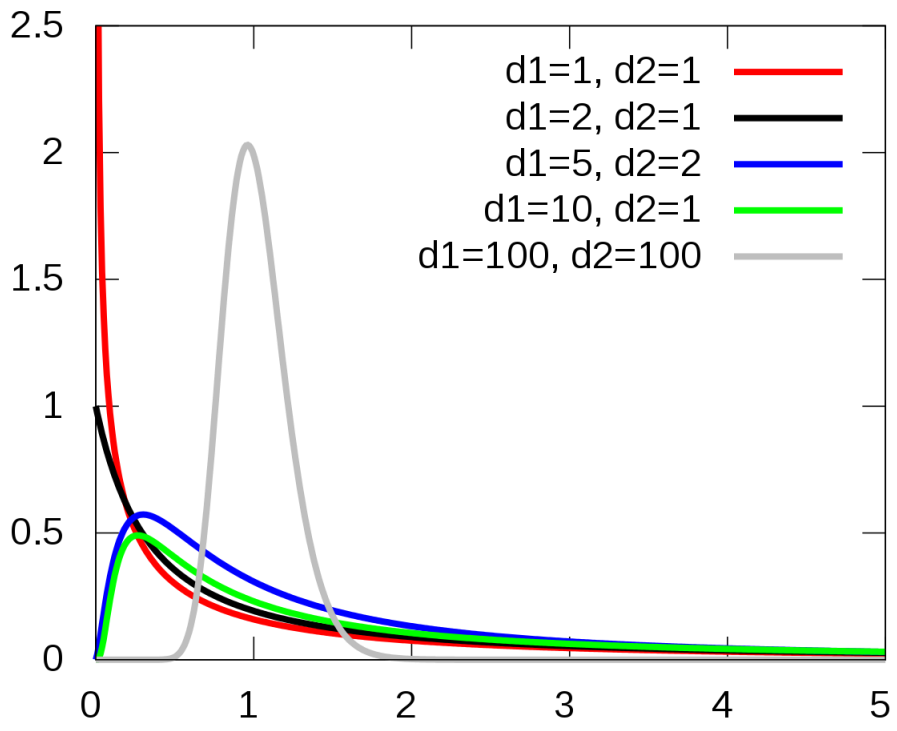
3. Plot the null distribution

If a few conditions (assumptions) are met, the null distribution for our F-statistic will be an F-distribution

The F-distribution is a family of distributions that have two parameters: df_1 and df_2

When using the F-distribution as a null distribution for our F-statistic, the appropriate parameter values are:

- $df_1 = K - 1$
- $df_2 = N - K$



3. Plot the null distribution

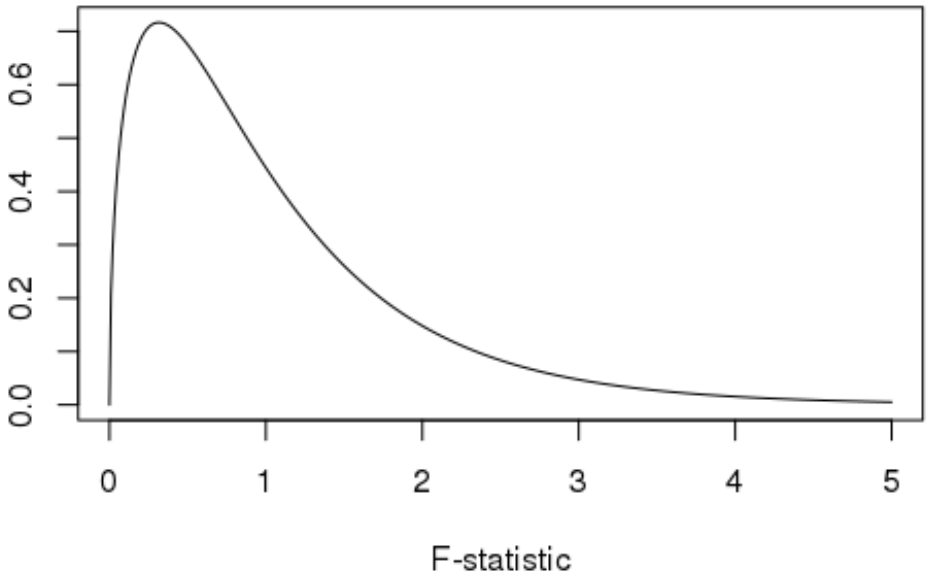
In R we can plot the density function for an F-distribution using the function…

- df(x_vals, df1, df2)

When using the F-distribution as a null distribution for our F-statistic, the appropriate parameter values are:

- $df_1 = K - 1 = 4 - 1 = 3$
- $df_2 = N - K = 40 - 4 = 36$

```
x_vals <- seq(0, 5, length.out = 1000)
y_vals <- df(x_vals, 3, 36)
plot(x_vals, y_vals, type = 'l')
```



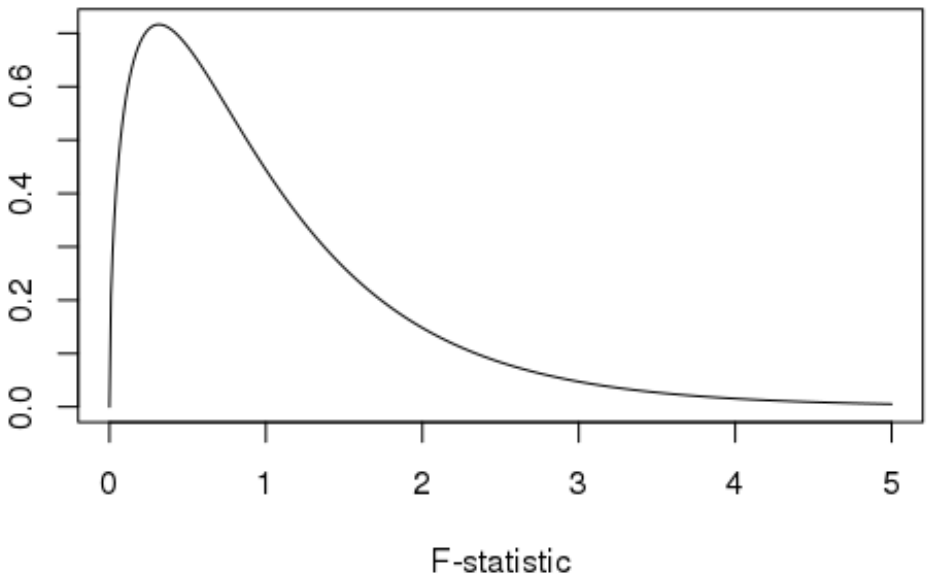
4. Calculate the p-value

In R we can plot the density function for an F-distribution using the function…

- df(x_vals, df1, df2)

When using the F-distribution as a null distribution for our F-statistic, the appropriate parameter values are:

- $df_1 = K - 1 = 4 - 1 = 3$
- $df_2 = N - K = 40 - 4 = 36$



abline(v = obs_stat, col = "red")

pf(obs_stat, 3, 36, lower.tail = FALSE)

= 0.267433

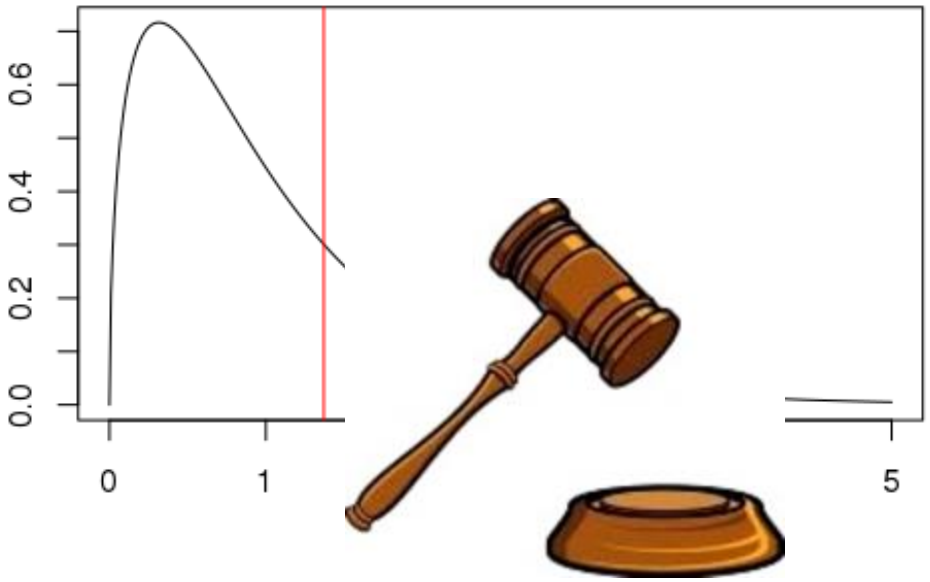
5. Conclusion?

In R we can plot the density function for an F-distribution using the function…

- df(x_vals, df1, df2)

When using the F-distribution as a null distribution for our F-statistic, the appropriate parameter values are:

- $df_1 = K - 1 = 4 - 1 = 3$
- $df_2 = N - K = 40 - 4 = 36$



abline(v = obs_stat, col = "red")

pf(obs_stat, 3, 36, lower.tail = FALSE)

= 0.267433

Brief mention/review: Permutation test comparing multiple means using the F-statistic

Sudoku by field

1. State the null and alternative hypotheses!

$\mathbf{H_0}$: $\mu_{as} = \mu_{ns} = \mu_{ss} = \mu_{ah}$

$\mathbf{H_A}$: $\mu_i \neq \mu_j$ for one pair of fields of study

Thoughts on the statistic of interest?

Comparing multiple means

There are many possible statistics we could use. A few choices are:

1. <u>Group range statistic:</u>

   $\max \bar{x} - \min \bar{x}$

2. <u>Mean absolute difference (MAD):</u>

   $(|\bar{x}_{as} - \bar{x}_{ns}| + |\bar{x}_{as} - \bar{x}_{ss}| + |\bar{x}_{as} - \bar{x}_{ah}| + |\bar{x}_{ns} - \bar{x}_{ss}| + |\bar{x}_{ns} - \bar{x}_{ah}| + |\bar{x}_{ss} - \bar{x}_{ah}|)/6$

3. <u>F statistic:</u>

$$F = \frac{\text{between-group variability}}{\text{within-group variability}}$$

The MAD statistic vs. the F-Statistic

<u>Mean absolute difference (MAD):</u>

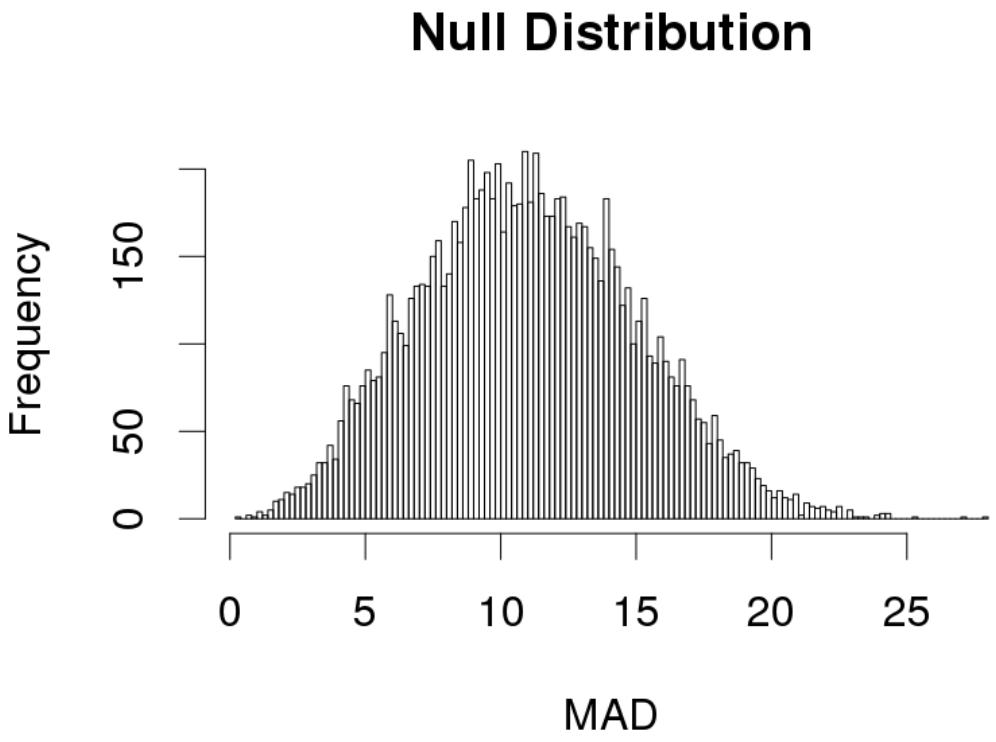
$(|\bar{x}_{as} - \bar{x}_{ns}| + |\bar{x}_{as} - \bar{x}_{ss}| + |\bar{x}_{as} - \bar{x}_{ah}| + |\bar{x}_{ns} - \bar{x}_{ss}| + |\bar{x}_{ns} - \bar{x}_{ah}| + |\bar{x}_{ss} - \bar{x}_{ah}|)/6$

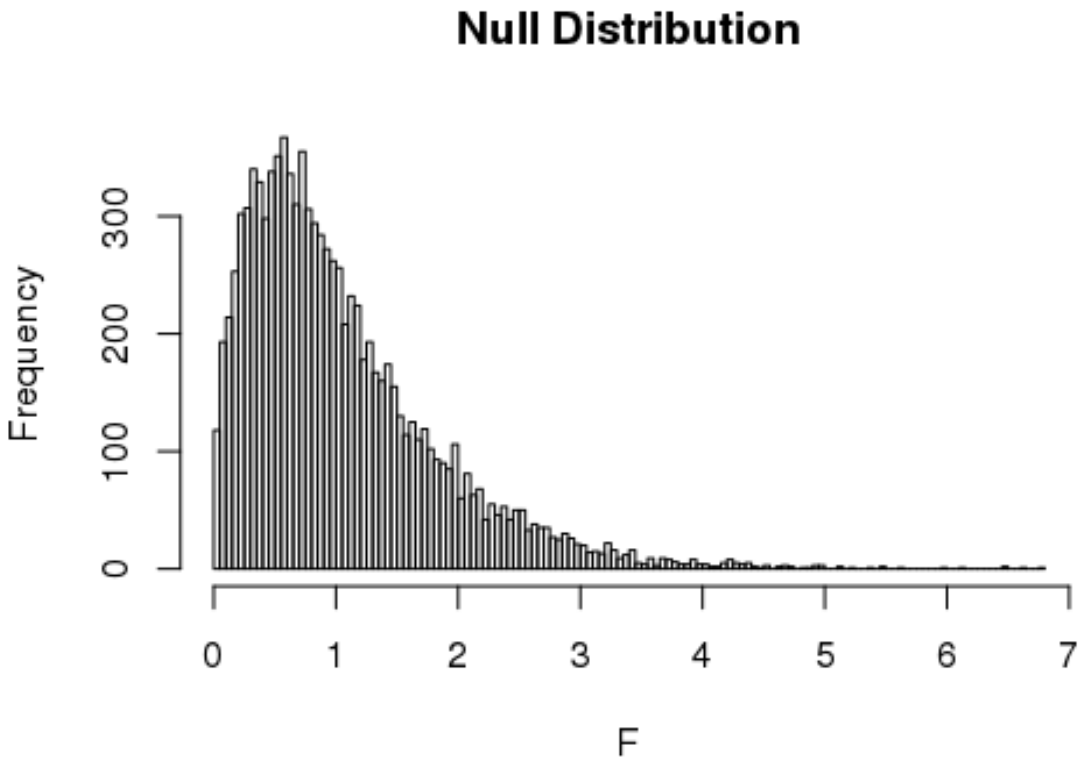
Observed MAD statistic value = 13.92

$$F = \frac{\text{between-group variability}}{\text{within-group variability}} = \frac{\frac{1}{K-1}\sum_{i=1}^{K} n_i(\bar{x}_i - \bar{x}_{tot})^2}{\frac{1}{N-K}\sum_{i=1}^{K}\sum_{j=1}^{n_i}(x_{ij} - \bar{x}_i)^2}$$

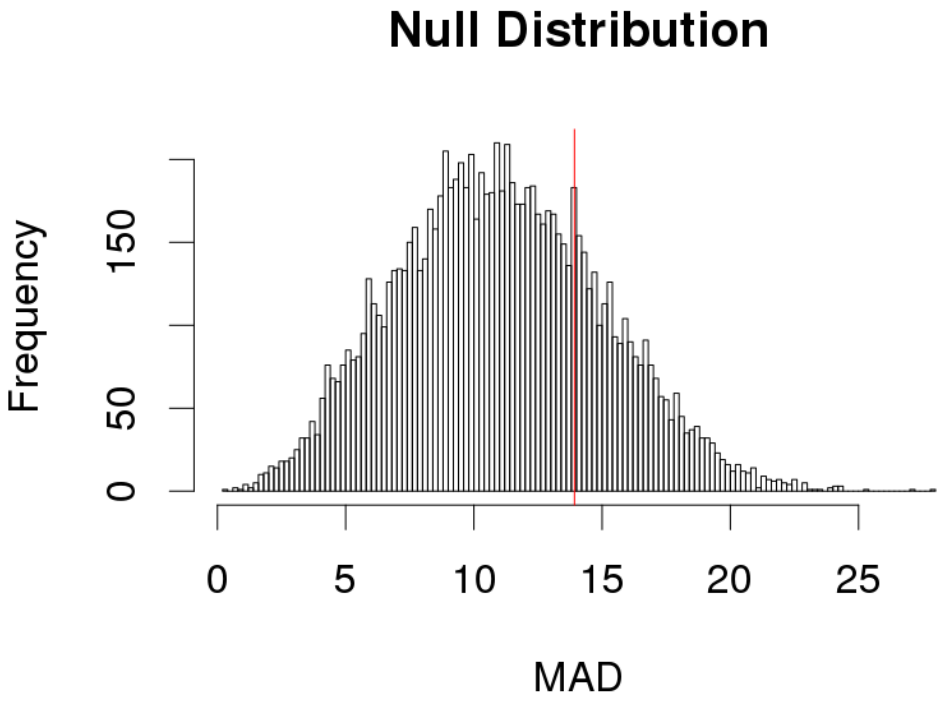
Observed F statistic value = 1.370

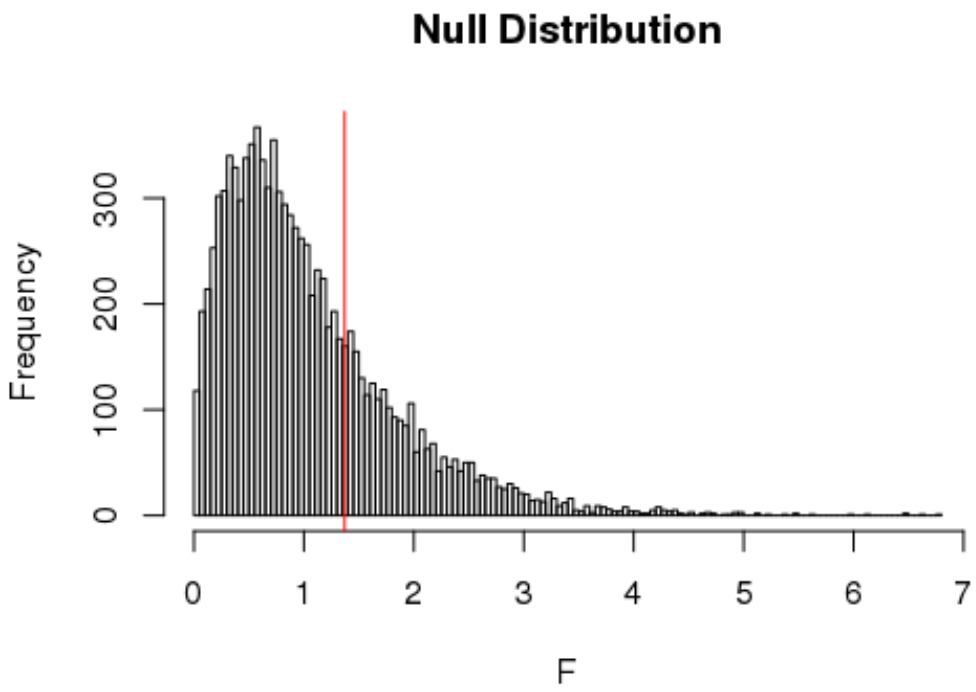
Null distributions





P-value

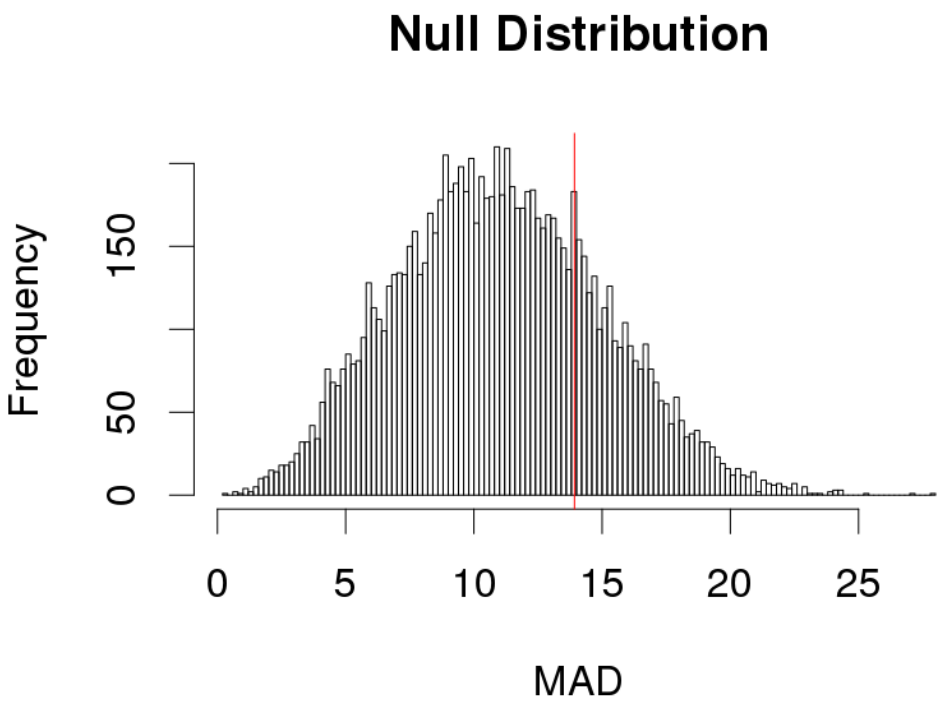


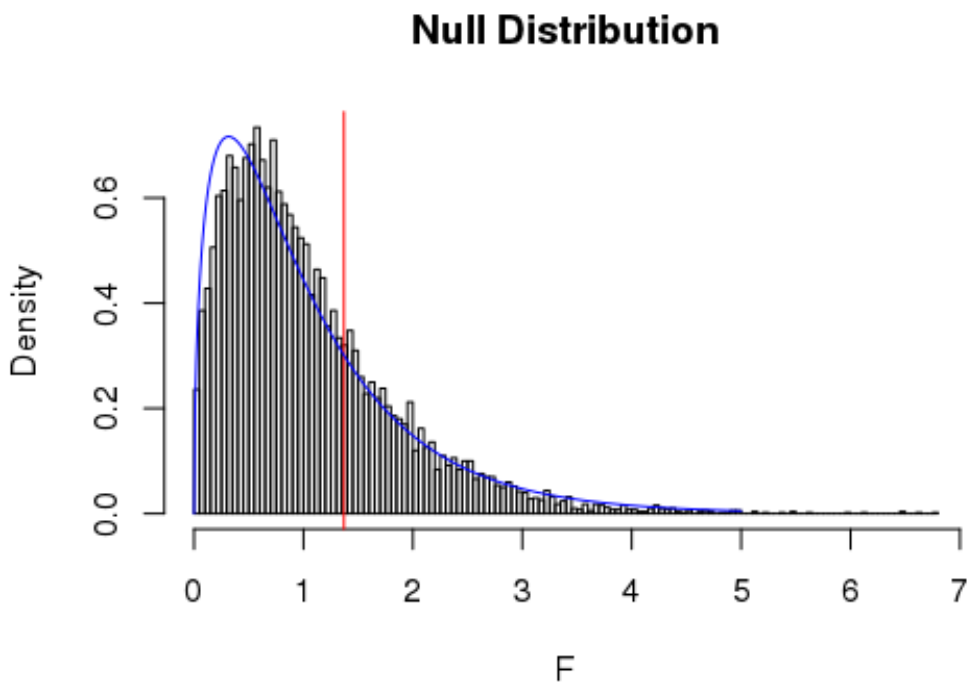


p-value = .4682

p-value = 0.2653

P-value





p-value = .4682

p-value = 0.2674

Conclusions?